"Calibrating a Simpson 260 VOM"

Disclaimers

- I am NOT a metrologist!
- I don't have "traceable" standards
- My methods are NOT certified
- I am only showing the voltage adjustments (not current shunts)
- I am not taking great care about meter orientation

My Intentions

- Explain & Demonstrate what each adjustment does
- Applies to Series 4 and up

THE SIMPSON 260 RANGE ACCURACY DEPENDS ON:

- THE METER MOVEMENT HAVING *1
  FULL SCALE = 50μA

- THE CORE METER CIRCUIT RESISTANCE *2
  IN THE 50μA FULL SCALE RANGE
  EQUAL TO 5000Ω.

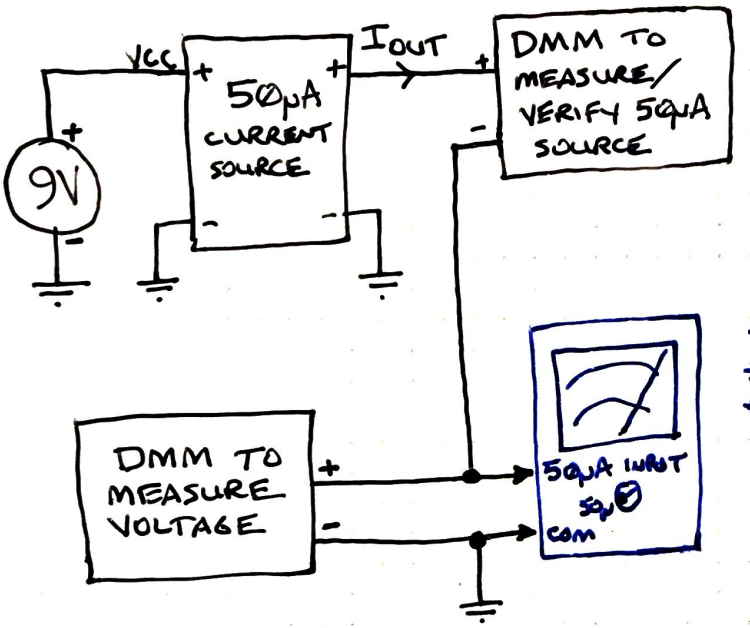
- PRECISION RESISTORS USED IN SERIES *3
  TO ESTABLISH EACH VOLTAGE RANGE

- FOR AC RANGES, ADDITIONAL SHUNT *4
  RESISTORS ARE USED

NOTES:

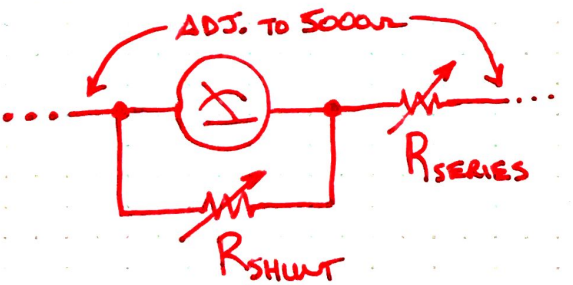
1) NOT ADJUSTABLE ON SERIES 3
2) ADJUSTABLE ON SERIES 3-8
3) NOT ADJUSTABLE
4) TWO ADJUSTMENTS (LOW & HIGH RANGE)

ADJUSTING 50μA FULL SCALE AND 5000Ω CORE RESISTANCE



BEFORE YOU START

- ZERO THE METER !!



STEPS

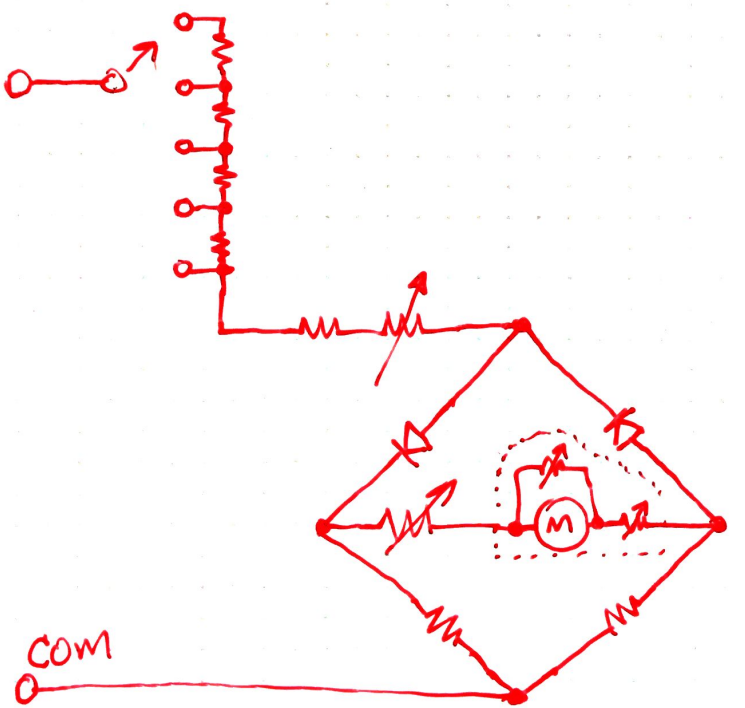
1) VERIFY 50μA CURRENT
2) ADJUST R_{SHUNT} FOR FULL SCALE
3) ADJUST R_{SERIES} FOR 250mV ON DMM

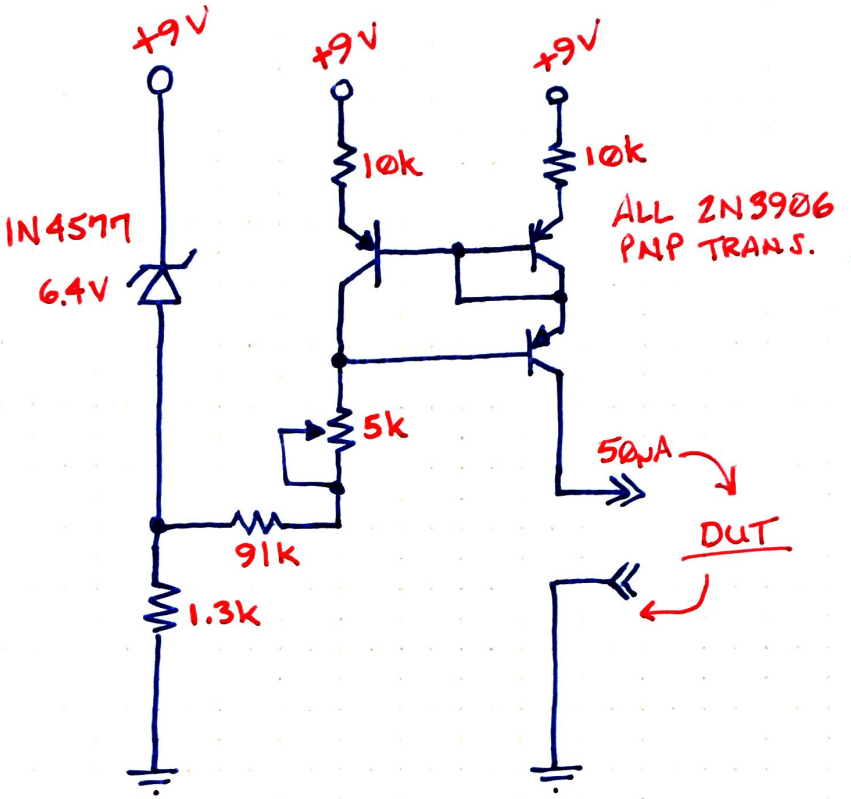
ADJUST AC VOLTAGE RANGES

(SEE DIAGRAM IN "THEORY OF OPERATION")

1) ADJUST RESISTOR "INSIDE" THE BRIDGE ON 250 VAC RANGE

2) ADJUST SERIES MULTIPLIER RESISTOR ON 2.5VAC RANGE



W2AEW

My 50μA Current Source